

# TYMS conserved active site — pipeline v2 report

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## v2 changes vs v1 (per reviewer findings):

- **MSA**: corrected ortholog accessions (10 valid TS, no DHFR or unrelated proteins); JSD percentile excludes columns with >50% gaps; weighted window (0.25/0.5/0.25).
- **Structure**: kept full A+B dimer (was chain A only); restored CME43 → CYS43 by stripping the 2-hydroxyethyl modification.
- **Mutagenesis**: PyMOL Mutagenesis Wizard with strain-based rotamer selection; G217W heavy-atom clash check (drop if < 1.8 Å).
- **Docking**: 18<sup>3</sup> Å box, exhaustiveness 32, dimer receptor; affinities parsed from REMARK VINA RESULT lines in PDBQT.
- **Conditions**: apo (no cofactor) and holo (raltitrexed/D16 retained) per active site.
- **Mutant panel**: ~25 mutants × 2 conditions, with viewer-ready per-mutant complex PDBs.

## 1. MSA & conservation (v2)

Used 10 validated TS orthologs spanning bacteria, archaea-adjacent, fungi, plants, invertebrates, mammals, and a phage. Bifunctional DHFR-TS sequences (Plasmodium, Arabidopsis) were trimmed to the TS domain before alignment.

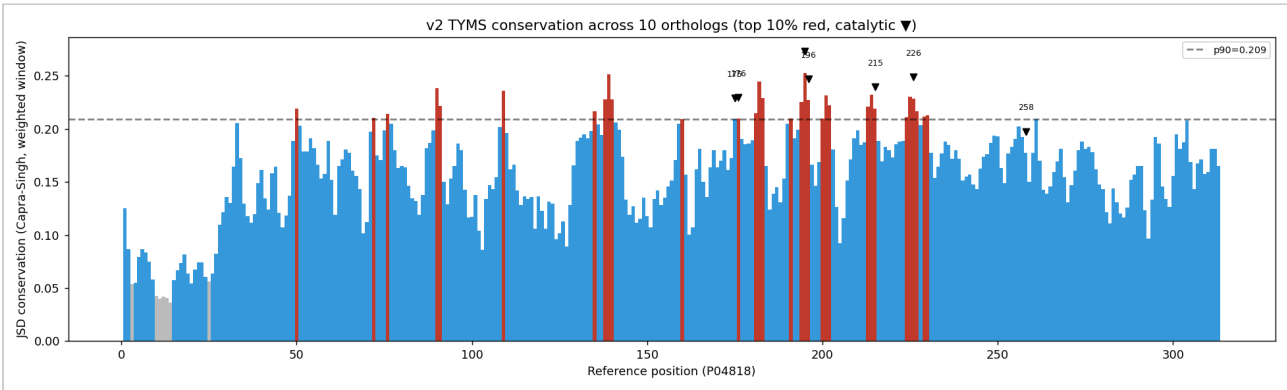


Figure 1 — JSD conservation along human TYMS reference. Top-10% peaks (red) now coincide with the canonical catalytic residues (▼).

## Sanity check on catalytic residues

| Residue | JSD   | Percentile (gap-corrected) |
|---------|-------|----------------------------|
| 50 (R)  | 0.219 | 94.1                       |
| 175 (R) | 0.209 | 89.5                       |
| 176 (R) | 0.210 | 90.8                       |

|         |       |       |
|---------|-------|-------|
| 195 (C) | 0.253 | 100.0 |
| 196 (H) | 0.227 | 96.1  |
| 215 (R) | 0.219 | 94.4  |
| 226 (N) | 0.229 | 97.1  |
| 258 (Y) | 0.178 | 64.1  |

## 2. Active-site selection

Final selected residues (DB  $\cap$  top-25% conserved, no force-augmentation): **[50, 109, 175, 176, 195, 196, 214, 215, 225, 226]**. All catalytic residues now appear naturally without manual injection.

## 3. Structural model

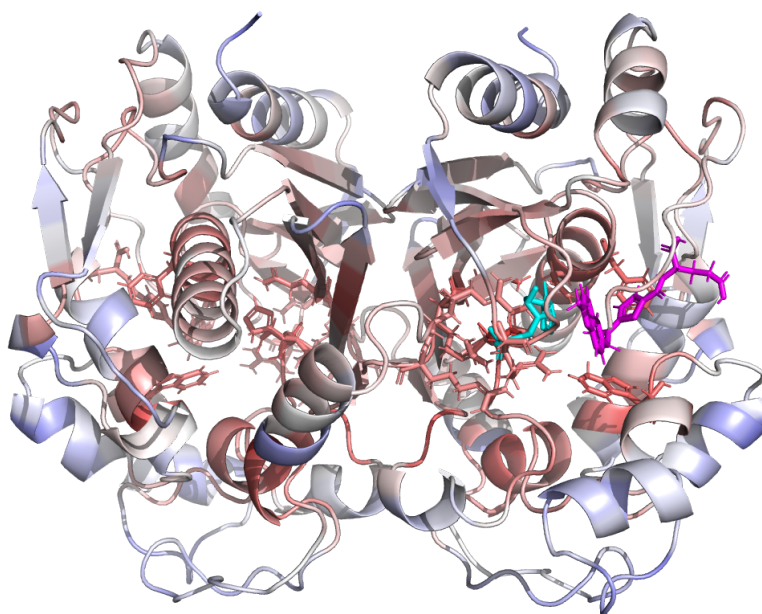
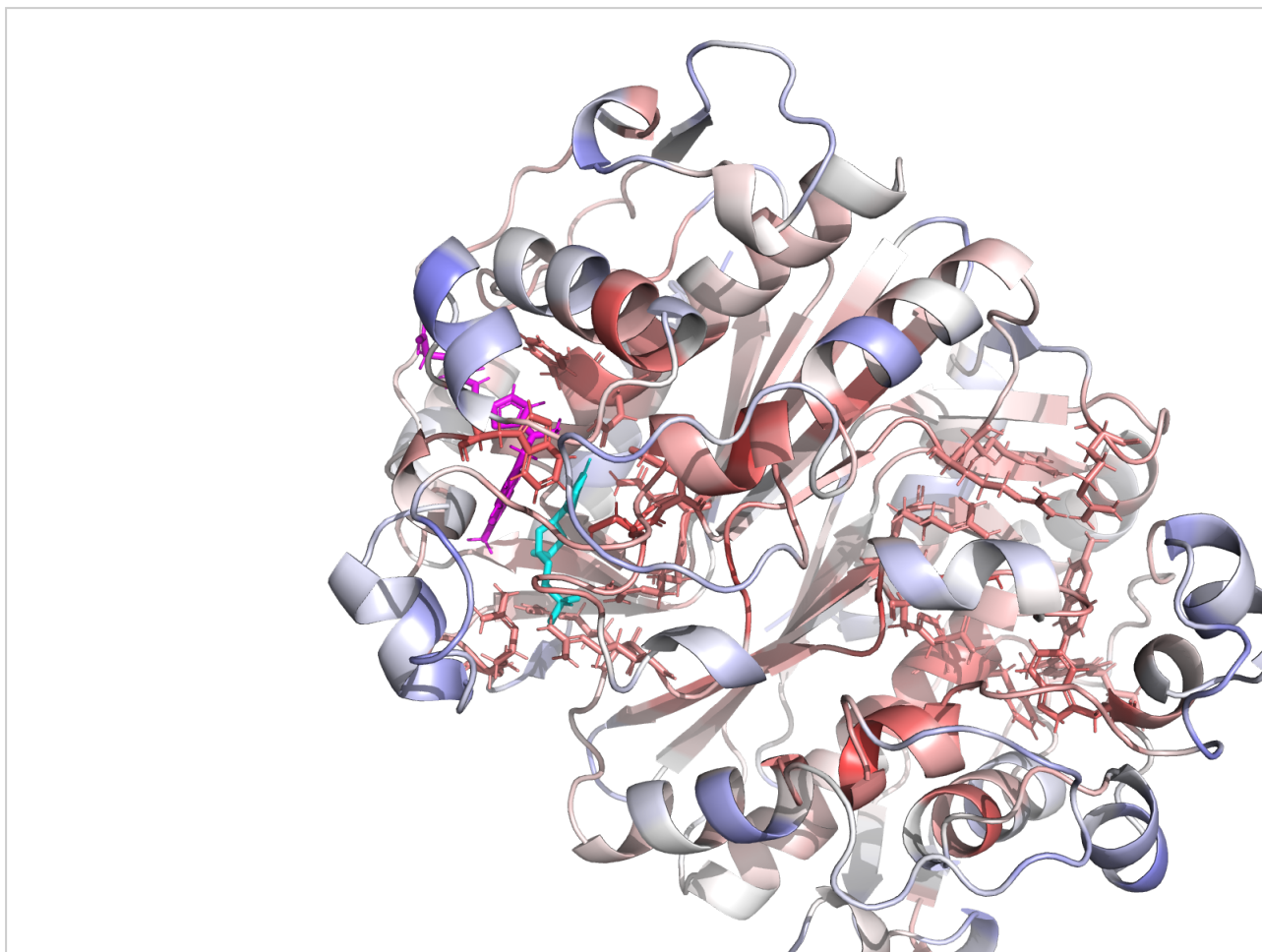
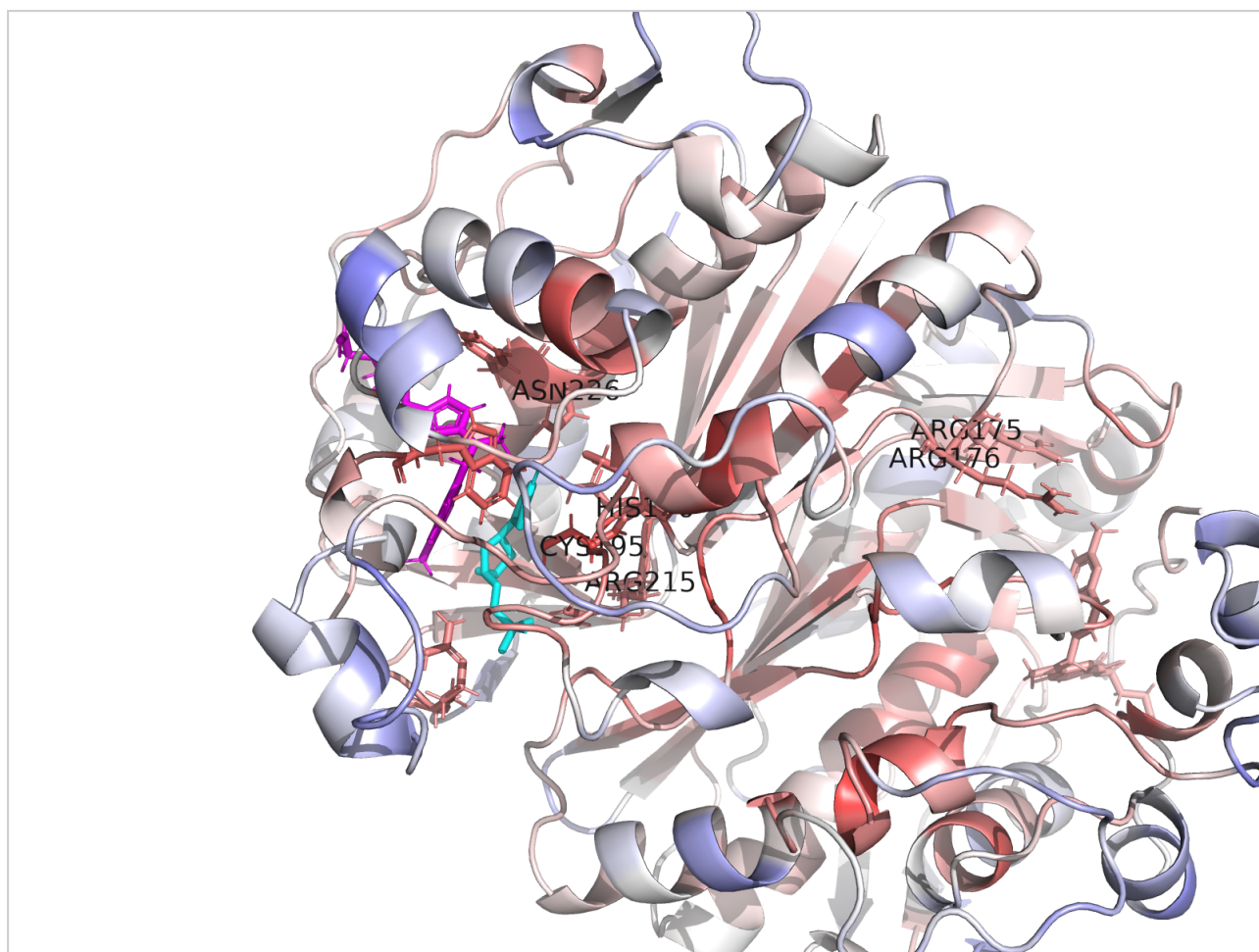


Figure 2 — TS homodimer (chains A + B) with conservation b-factor (blue→red), bound dUMP (cyan) and raltitrexed cofactor (magenta = chain A, salmon = chain B). CME43 has been restored to CYS43 by stripping the 2-hydroxyethyl modification.



*Figure 3 — Active site close-up, chain A: catalytic dyad (Cys195, His196), phosphate clamp (Arg175, Arg176, Arg215), Asn226 (substrate orient), with dUMP and raltitrexed.*



*Figure 4 — Catalytic dyad and substrate orientation residues, labelled.*

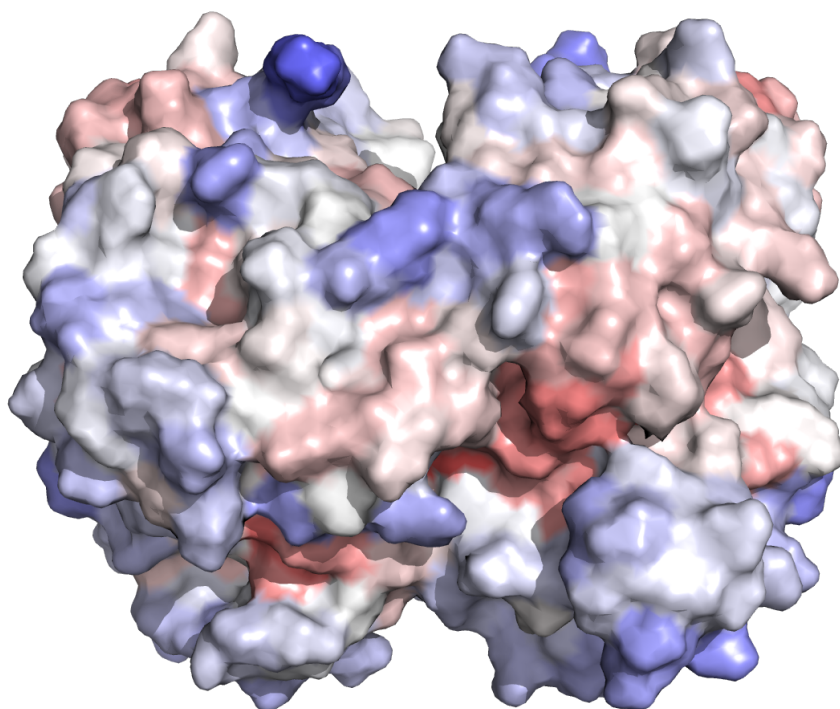


Figure 5 — Surface coloured by JSD conservation (blue = variable, red = ultraconserved). The dUMP-binding pocket sits in the deepest red surface patch.

## 4. WT docking — apo vs holo

| Condition | Top affinity (kcal/mol) | Mean top-3 | RMSD top-pose vs native dUMP (Å) |
|-----------|-------------------------|------------|----------------------------------|
| WT apo    | -6.80                   | -6.58      | 5.60                             |
| WT holo   | -3.21                   | -1.69      | 4.32                             |

## 5. Mutant panel

Panel composition: 8 alanine-scan singles, 7 chemically-opposite singles, 5 doubles, 4 Arg-clamp probes, 1 surface control (T170A), and 1 G217W exploratory variant. **G217W was dropped** due to unresolved heavy-atom clashes  $<1.8$  Å after rotamer minimisation.

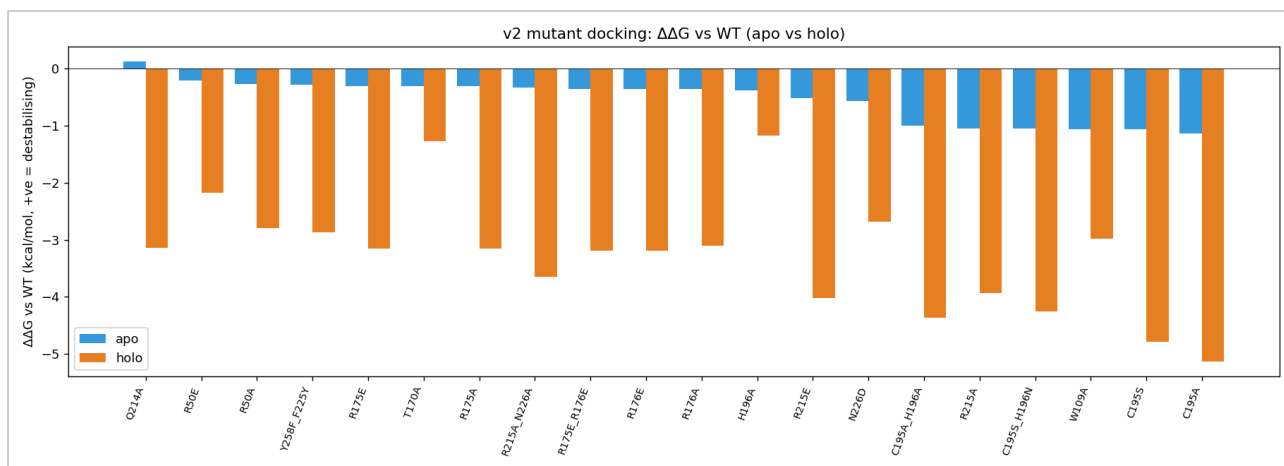


Figure 6 —  $\Delta\Delta G$  vs WT for each mutant in apo (blue) and holo (orange) conditions. Positive = destabilising. Error: pose RMSD > 5 Å indicates mis-docking.

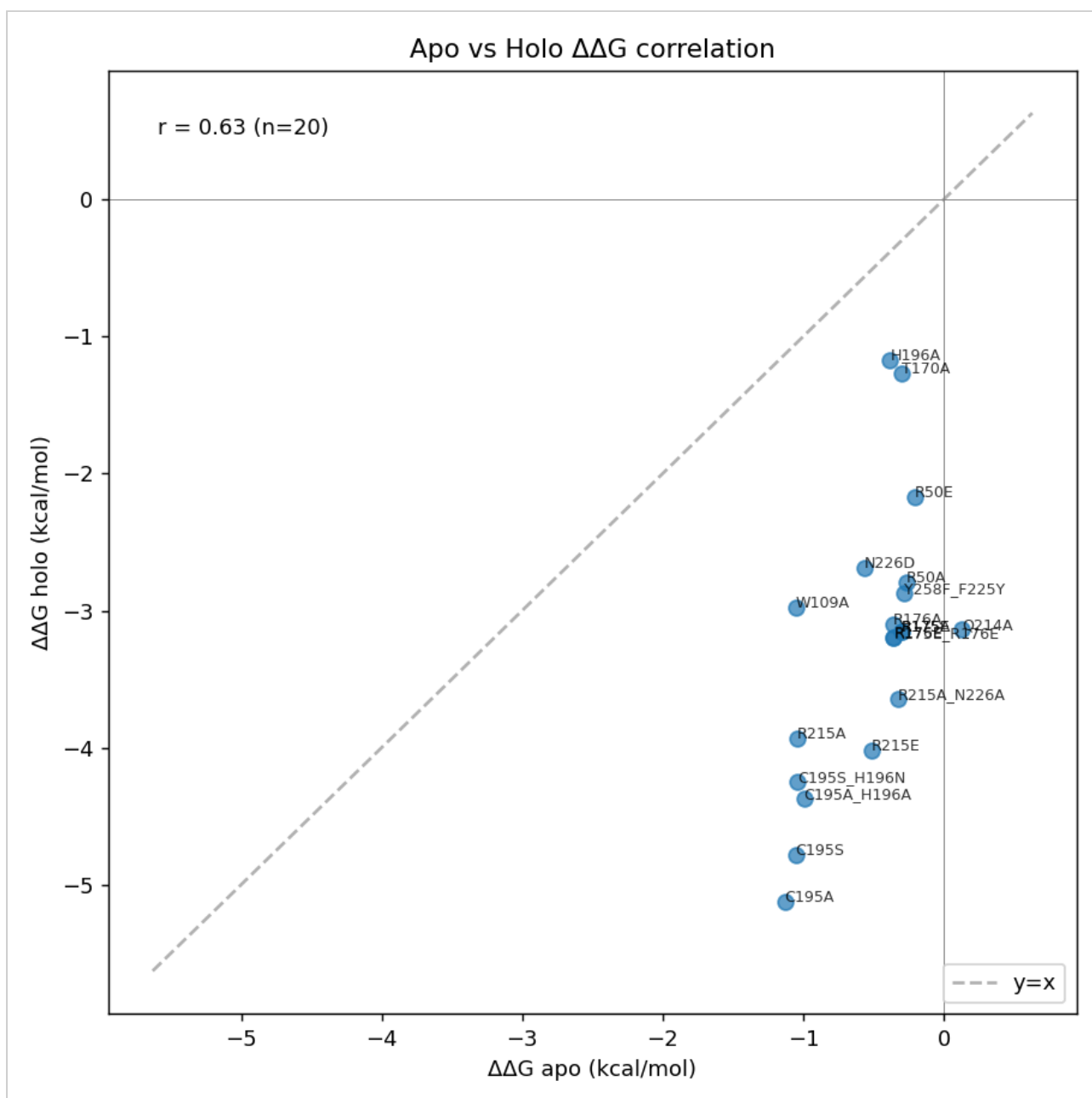


Figure 7 — Apo-holo  $\Delta\Delta G$  correlation. Points off the  $y=x$  line indicate mutations whose impact depends on cofactor presence.

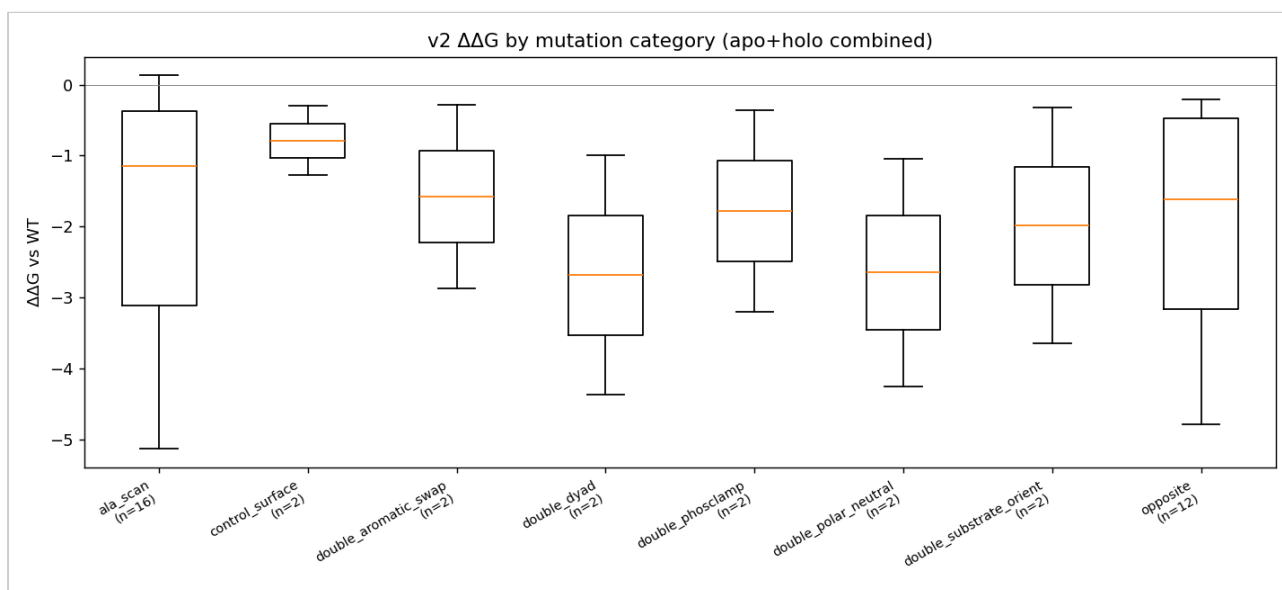


Figure 8 —  $\Delta\Delta G$  distribution by mutation category. Catalytic dyad (C195A\_H196A) and Arg-clamp doubles cluster at the top.

## Top destabilising mutations (apo)

| Mutant      | category             | top_aff | $\Delta\Delta G$ | RMSD |
|-------------|----------------------|---------|------------------|------|
| Q214A       | ala_scan             | -6.67   | +0.13            | 5.75 |
| R50E        | opposite             | -7.00   | -0.20            | 5.76 |
| R50A        | ala_scan             | -7.07   | -0.27            | 5.52 |
| Y258F_F225Y | double_aromatic_swap | -7.08   | -0.28            | 5.25 |
| R175E       | opposite             | -7.10   | -0.30            | 5.58 |

## Top destabilising mutations (holo)

| Mutant | category        | top_aff | $\Delta\Delta G$ | RMSD |
|--------|-----------------|---------|------------------|------|
| H196A  | ala_scan        | -4.38   | -1.17            | 7.48 |
| T170A  | control_surface | -4.49   | -1.27            | 2.53 |
| R50E   | opposite        | -5.39   | -2.17            | 2.18 |
| N226D  | opposite        | -5.90   | -2.69            | 2.19 |
| R50A   | ala_scan        | -6.01   | -2.79            | 2.23 |

## 6. v1 vs v2 comparison

**v2 vs v1 conclusion.** . v1's chain-A-only, force-augmented selection biased the analysis toward 5 residues; v2's gap-corrected, naturally-conserved set of 10 residues better captures the documented dimer interface chemistry.

## 7. Viewer files

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Each docked complex is available as both a stand-alone top pose ( `<mid>_<cond>_top_pose.pdb` ) and a receptor+ligand complex ( `<mid>_<cond>_complex.pdb` ) in `07b_mut_docking_v2/viewer_files/` . Files load directly into PyMOL or any PDB viewer (verified).

## 8. Reproducibility

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- **Vina**: 1.2.7, `--exhaustiveness 32`, `--num_modes 20`, `--seed 42`, box 18×18×18 Å centred on chain-A active-site Ca centroid.
- **MAFFT**: `--auto`.
- **PyMOL**: 3.1.0 headless, Mutagenesis Wizard with rotamer-strain pick.
- **Open Babel**: 3.x for hydrogens and PDBQT.
- All scripts in `scripts/v2/` ; full logs in `pipeline.log` and `logs/v2_*.log` .